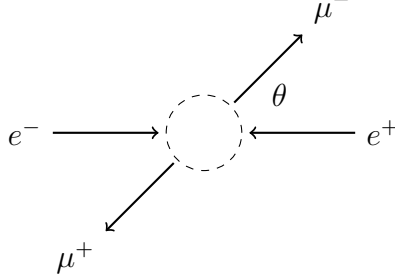


Muon pair production

Muon pair production is the process $e^- + e^+ \rightarrow \mu^- + \mu^+$.



The following center-of-mass momentum vectors have $p = \sqrt{E^2 - m^2}$ and $\rho = \sqrt{E^2 - M^2}$.

$$p_1 = \begin{pmatrix} E \\ 0 \\ 0 \\ p \end{pmatrix}_{e^- \rightarrow} \quad p_2 = \begin{pmatrix} E \\ 0 \\ 0 \\ -p \end{pmatrix}_{\leftarrow e^+} \quad p_3 = \begin{pmatrix} E \\ \rho \sin \theta \cos \phi \\ \rho \sin \theta \sin \phi \\ \rho \cos \theta \end{pmatrix}_{\nearrow \mu^-} \quad p_4 = \begin{pmatrix} E \\ -\rho \sin \theta \cos \phi \\ -\rho \sin \theta \sin \phi \\ -\rho \cos \theta \end{pmatrix}_{\nwarrow \mu^+}$$

Spinors for p_1 .

$$u_{11} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} E+m \\ 0 \\ p \\ 0 \end{pmatrix}_{\text{spin up}} \quad u_{12} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} 0 \\ E+m \\ 0 \\ -p \end{pmatrix}_{\text{spin down}}$$

Spinors for p_2 .

$$v_{21} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} -p \\ 0 \\ E+m \\ 0 \end{pmatrix}_{\text{spin up}} \quad v_{22} = \frac{1}{\sqrt{E+m}} \begin{pmatrix} 0 \\ p \\ 0 \\ E+m \end{pmatrix}_{\text{spin down}}$$

Spinors for p_3 .

$$u_{31} = \frac{1}{\sqrt{E+M}} \begin{pmatrix} E+M \\ 0 \\ p_{3z} \\ p_{3x} + ip_{3y} \end{pmatrix}_{\text{spin up}} \quad u_{32} = \frac{1}{\sqrt{E+M}} \begin{pmatrix} 0 \\ E+M \\ p_{3x} - ip_{3y} \\ -p_{3z} \end{pmatrix}_{\text{spin down}}$$

Spinors for p_4 .

$$v_{41} = \frac{1}{\sqrt{E+M}} \begin{pmatrix} p_{4z} \\ p_{4x} + ip_{4y} \\ E+M \\ 0 \end{pmatrix}_{\text{spin up}} \quad v_{42} = \frac{1}{\sqrt{E+M}} \begin{pmatrix} p_{4x} - ip_{4y} \\ -p_{4z} \\ 0 \\ E+M \end{pmatrix}_{\text{spin down}}$$

The scattering amplitude $\mathcal{M}_{abcd}$ for spin state $abcd$ is

$$\mathcal{M}_{abcd} = \frac{e^2}{s} (\bar{u}_{3c} \gamma^\mu v_{4d}) (\bar{v}_{2b} \gamma_\mu u_{1a})$$

Symbol s is the Mandelstam variable

$$s = (p_1 + p_2)^2 = 4E^2$$

In component form

$$\mathcal{M}_{abcd} = \frac{e^2}{s} [(\bar{u}_{3c})_\alpha \gamma^{\mu\alpha}{}_\beta (v_{4d})^\beta] [(\bar{v}_{2b})_\rho \gamma_\mu{}^\rho{}_\sigma (u_{1a})^\sigma]$$

The average density $\langle |\mathcal{M}|^2 \rangle$ is the sum over squared amplitudes divided by the number of inbound states.

$$\begin{aligned} \langle |\mathcal{M}|^2 \rangle &= \frac{1}{4} \sum_{abcd} |\mathcal{M}_{abcd}|^2 \\ &= \frac{e^4}{64E^4} \sum_{abcd} |(\bar{u}_{3c} \gamma^\mu v_{4d}) (\bar{v}_{2b} \gamma_\mu u_{1a})|^2 \end{aligned}$$

The Casimir trick uses matrix arithmetic to sum over spin states.

$$\langle |\mathcal{M}|^2 \rangle = \frac{e^4}{64E^4} \text{Tr} \left[(\not{p}_3 + M) \gamma^\mu (\not{p}_4 - M) \gamma^\nu \right] \text{Tr} \left[(\not{p}_2 - m) \gamma_\mu (\not{p}_1 + m) \gamma_\nu \right]$$

The result is

$$\langle |\mathcal{M}|^2 \rangle = e^4 \left(1 + \cos^2 \theta + \frac{m^2 + M^2}{E^2} \sin^2 \theta + \frac{m^2 M^2}{E^4} \cos^2 \theta \right)$$

The differential cross section is

$$\frac{d\sigma}{d\Omega} = \frac{\langle |\mathcal{M}|^2 \rangle}{4(4\pi\epsilon_0)^2 s}$$

where

$$s = (p_1 + p_2)^2 = 4E^2$$

For the high energy approximation

$$\langle |\mathcal{M}|^2 \rangle = e^4 (1 + \cos^2 \theta)$$

we have

$$\frac{d\sigma}{d\Omega} = \frac{e^4}{4(4\pi\epsilon_0)^2 s} (1 + \cos^2 \theta)$$

Noting that

$$e^2 = 4\pi\epsilon_0 \alpha \hbar c$$

we also have

$$\frac{d\sigma}{d\Omega} = \frac{(\alpha \hbar c)^2}{4s} (1 + \cos^2 \theta)$$